Real Time Operating System

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Rugged Board

Real Time Application (RTA)

Myth

The Real-Time Computing is equivalent to fast computing

The Real-Time Computing means higher performance

Truth

The Real-Time Computing is equivalent to Predictable computing

In Real-Time Computing timeliness is more important than performance.

"Real-time deals with guarantees, not with raw speed.

“A Real time system is one in which the correctness of the computations not only depends upon the logical correctness of the computation, but also upon the time at which the result is produced . If the timing constraints are not met, system failure is said to have occurred “

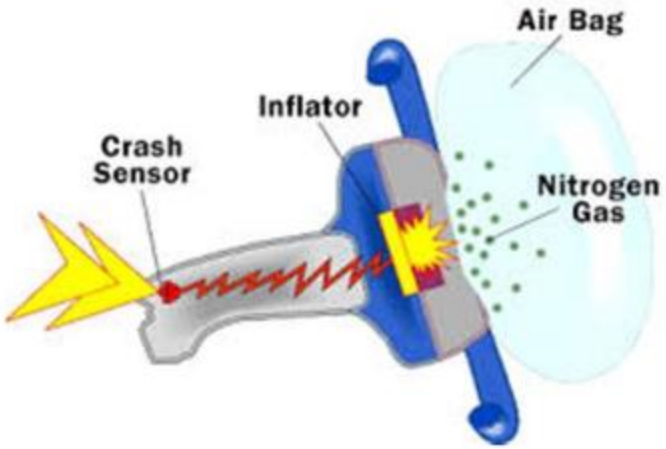
What is Real time ?

RTOS

What are Real time Applications(RTAs) ?

- RTAs are not fast executing applications.
- RTAs are time deterministic applications, that means, their response time to events is almost constant.
- There could be small deviation in RTAs response time, in terms of ms or seconds which will fall into the category of soft real time applications.
- Hard real time functions must complete within a given time limit. Failure to do so will result in absolute failure of the system

What are Real time Applications(RTAs) ?



Hard-Real Time application

May be soft-real time application ?
Delay is tolerable

RTOS

Real-Time Application

Time Deterministic – Response time to events is always almost constant .

Can always trust RTAs in terms of its timings in responding to events

RTOS

What is Real Time Operating system?

It’s an OS , specially designed to run applications with very precise timing and a high degree of reliability.
To be considered as "real-time", an operating system must have a known maximum time for each of the critical operations that it performs.
Some of these operations include

- Handling of interrupts and internal system exceptions
- Handling of Critical Sections.
- Scheduling Mechanism , etc.

RTOS vs GPOS: Task Scheduling

GPOS
In the case of a GPOS – task scheduling is not based on “priority” always!
RTOS
Where as in RTOS – scheduling is always priority based.
Most RTOS use pre-emptive task scheduling method which is based on priority levels.

GPOS is programmed to handle scheduling in such a way that it manages to achieve high throughput. Throughput means – the total number of processes that complete their execution per unit time
Some times execution of a high priority process will get delayed in order to serve 5 or 6 low priority tasks.
High throughput is achieved by serving 5 low priority tasks than by serving a single high priority one.

In a GPOS, the scheduler typically uses a fairness policy to dispatch threads and processes onto the CPU.

Such a policy enables the high overall throughput required by desktop and server applications, but offers no guarantees that high-priority, time critical threads or processes will execute in preference to lower-priority threads.

RTOS vs GPOS: Task Scheduling

RTOS- Task Scheduling
RTOS, Threads execute in the order of their priority.
If a high-priority thread becomes ready to run, it will take over the CPU from any lower-priority thread that may be executing.
Hear a high priority thread gets executed over the low priority ones. All “low priority thread execution” will get paused.
A high priority thread execution will get override only if a request comes from an even high priority threads.

Does that mean , RTOS are very poor in throughput ?

RTOS may yield less throughput than the General Purpose OS, because it always favors the high Priority task to execute first , but that does not mean, it has very poor throughput !
A Quality RTOS will still deliver decent overall throughput but can sacrifice throughput for being deterministic or to achieve time predictability .

RTOS vs GPOS: Task Scheduling

GPOS

Meet Higher Throughput

RTOS

Meet Time Predictability

For the RTOS, **achieving predictability or time deterministic nature is more important than throughput,** but for the GPOS achieving higher throughput for user convenience is more important

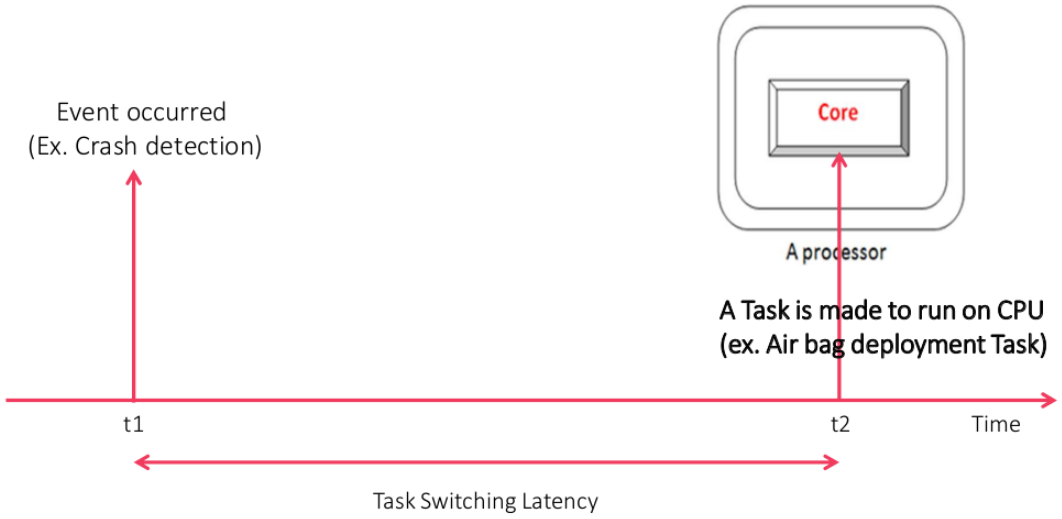
RTOS

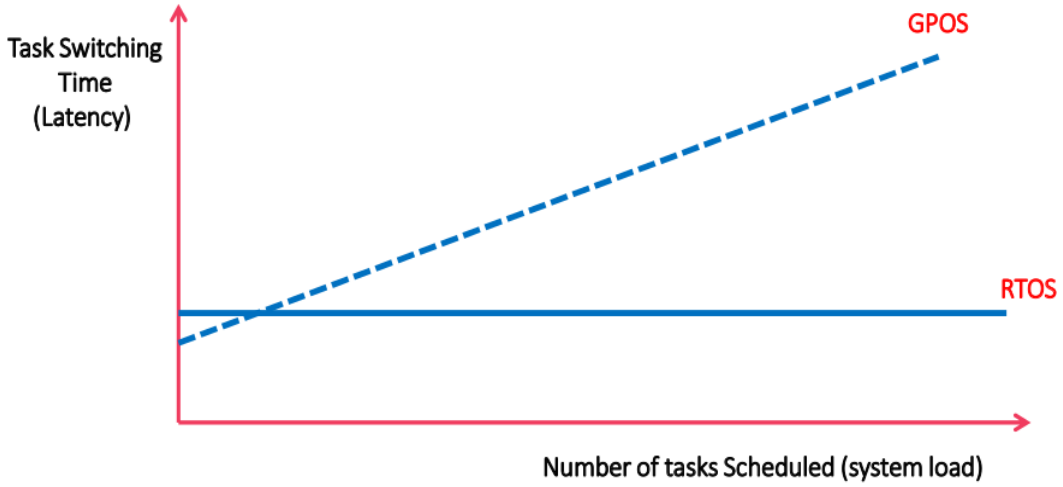
RTOS vs GPOS: Task Switching

Latency In Computing , Latency means “The time that elapses between a stimulus and the response to it”.

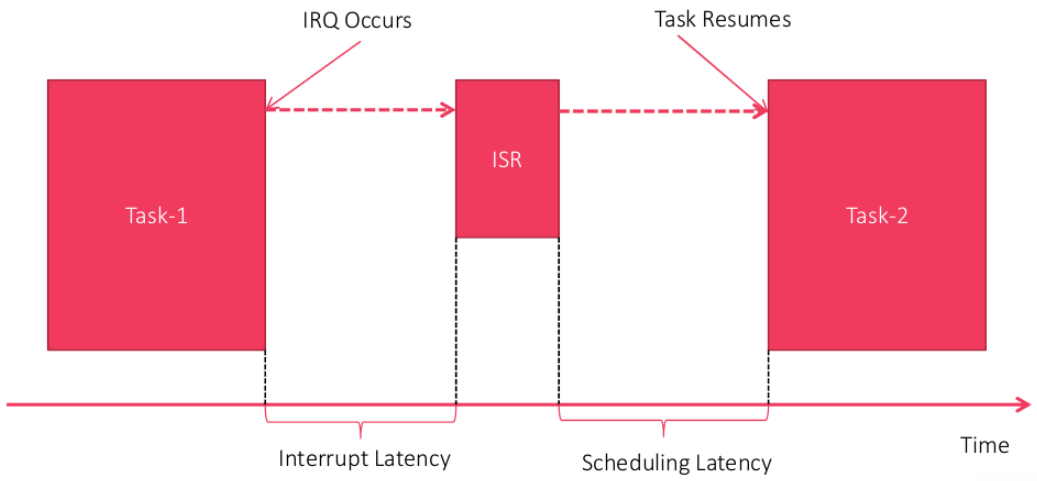
Task switching latency means, that time gap between

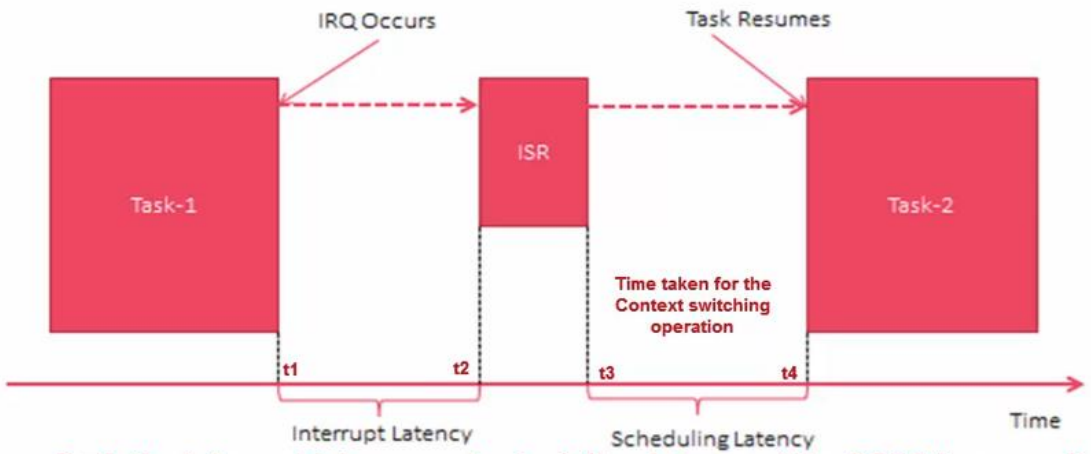
“A triggering of an event and the time at which the task which takes care of that event is allowed to run on the CPU”





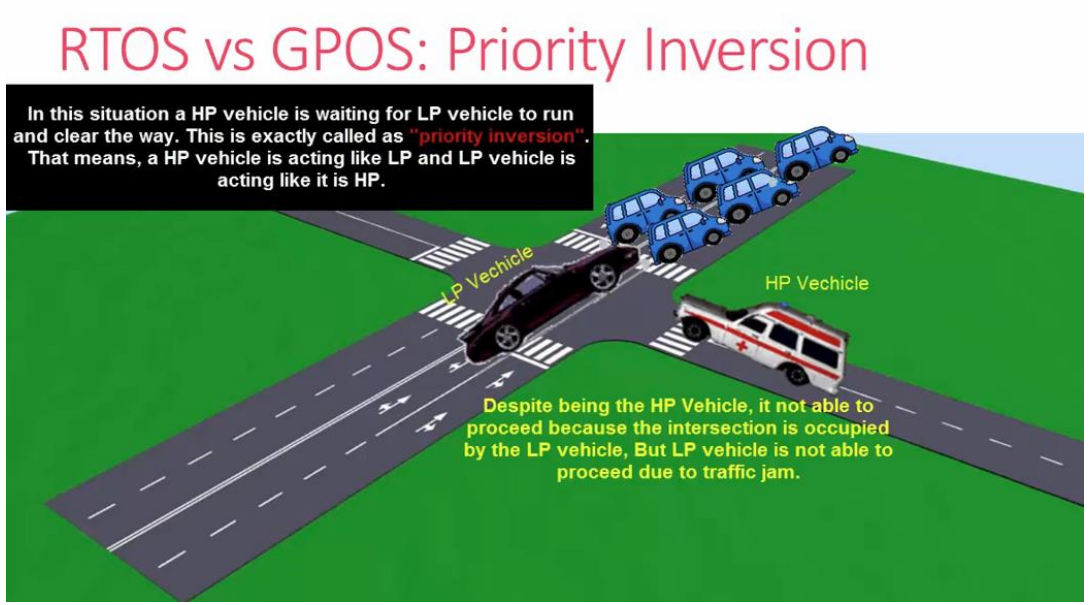
RTOS vs GPOS: Interrupt Latency

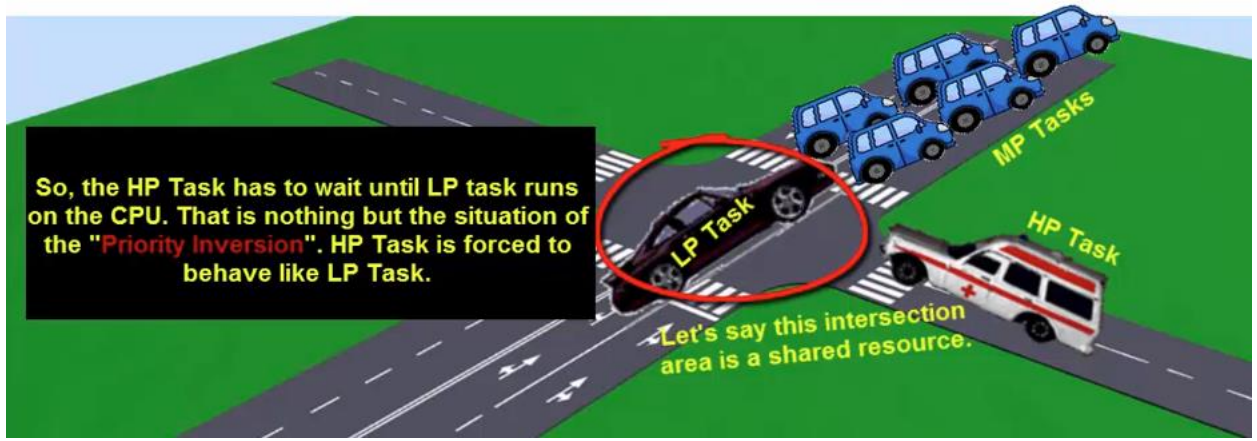


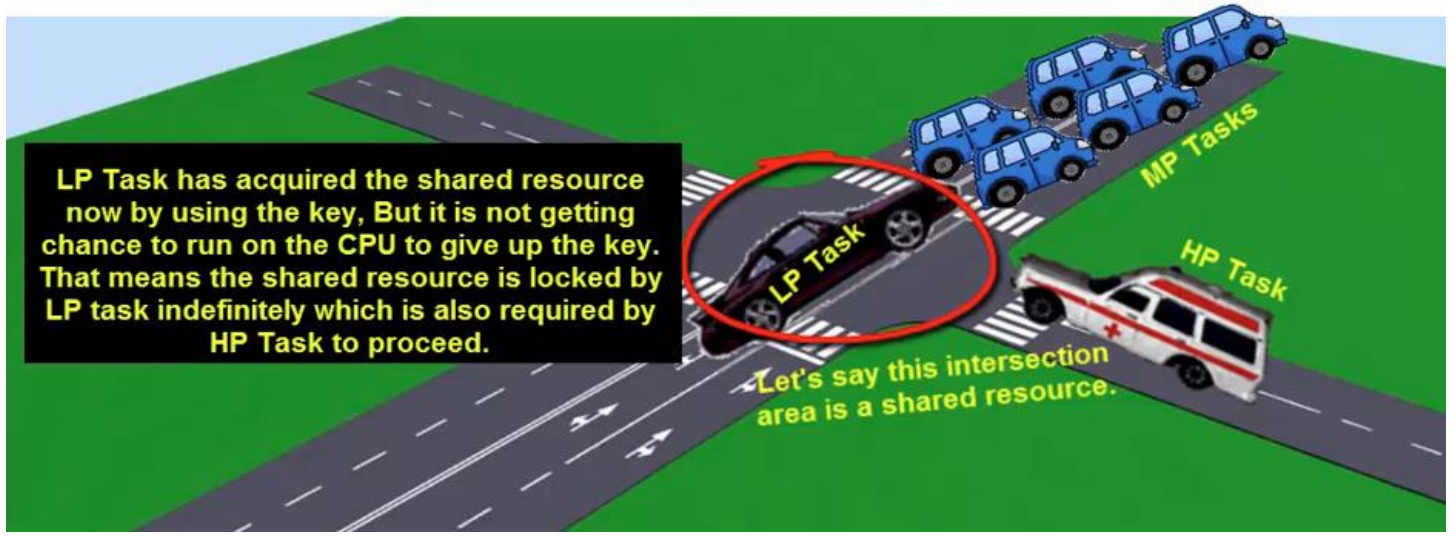


Both the interrupt latency and scheduling latency of the RTOS is as small as possible and time bounded. But in the case of GPOS, due to increase in system load these parameters may vary significantly.

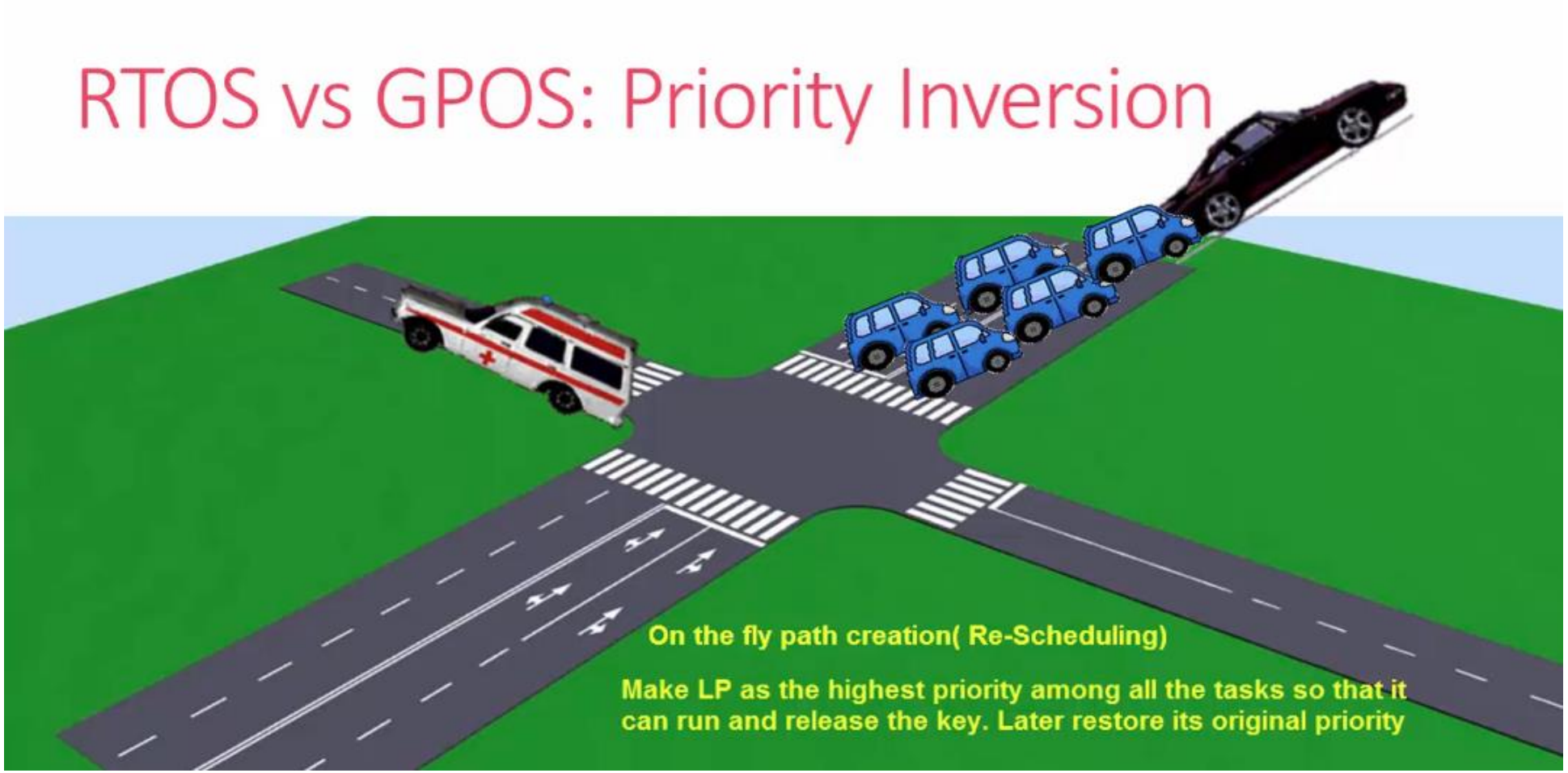
RTOS


RTOS vs GPOS: Priority Inversion
In this situation a HP vehicle is waiting for LP vehicle to run and clear the way. This is exactly called as "priority inversion". That means, a HP vehicle is acting like LP and LP vehicle is acting like it is HP.
LP Vechicle
HP Vechicle
Despite being the HP Vehicle, it not able to proceed because the intersection is occupied by the LP vehicle, But LP vehicle is not able to proceed due to traffic jam.


So, the HP Task has to wait until LP task runs on the CPU. That is nothing but the situation of the "Priority Inversion". HP Task is forced to behave like LP Task.
LP Task
MP Tasks
HP Task
Let's say this intersection area is a shared resource.


LP Task has acquired the shared resource now by using the key, But it is not getting chance to run on the CPU to give up the key. That means the shared resource is locked by LP task indefinitely which is also required by HP Task to proceed.
LP Task
MP Tasks
HP Task
Let's say this intersection area is a shared resource.

RTOS


RTOS vs GPOS: Priority Inversion
On the fly path creation(Re-Scheduling)
Make LP as the highest priority among all the tasks so that it can run and release the key. Later restore its original priority

RTOS vs GPOS: Priority Inversion

GPOS

Priority inversion effects are in-significant

RTOS

Priority inversion effects must be solved

RTOS

RTOS vs GPOS: Kernel Preemption

Preemption

In computing preemption is the act of temporarily removing the task from the running state without its co-operation

In RTOS , Threads execute in order of their priority.

f a high-priority thread becomes ready to run, it will, within a small and bounded time interval, take over the CPU from any lower-priority thread that may be executing that we call preemption.

The lower priority task will be made to leave CPU, if higher priority task wants to execute.

The kernel operations of a RTOS are pre-emptible where as the kernel operations of a GPOS are not pre-emptible

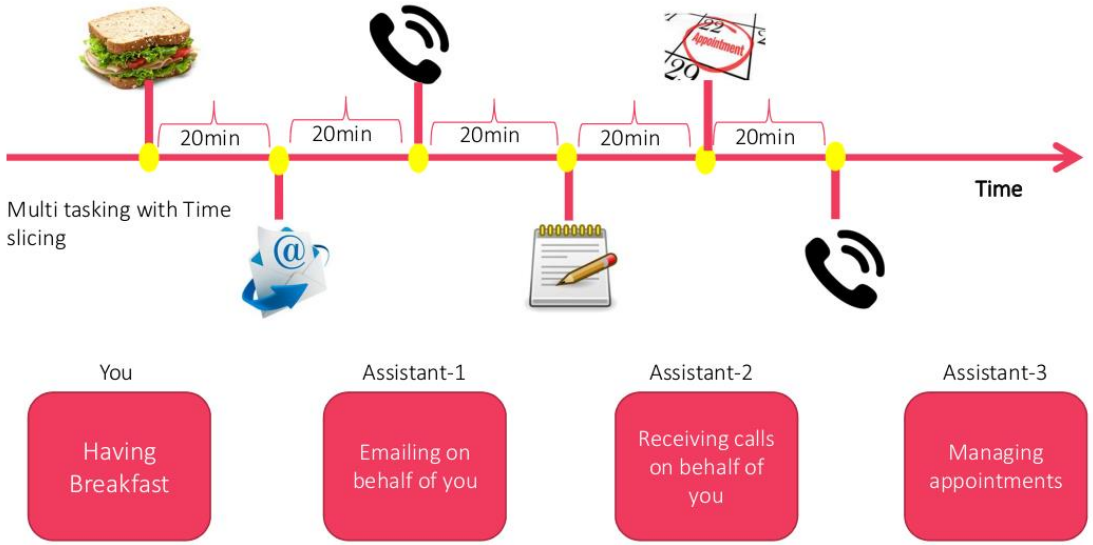
RTOS

What are the features that a RTOS has but a GPOS doesn't ?

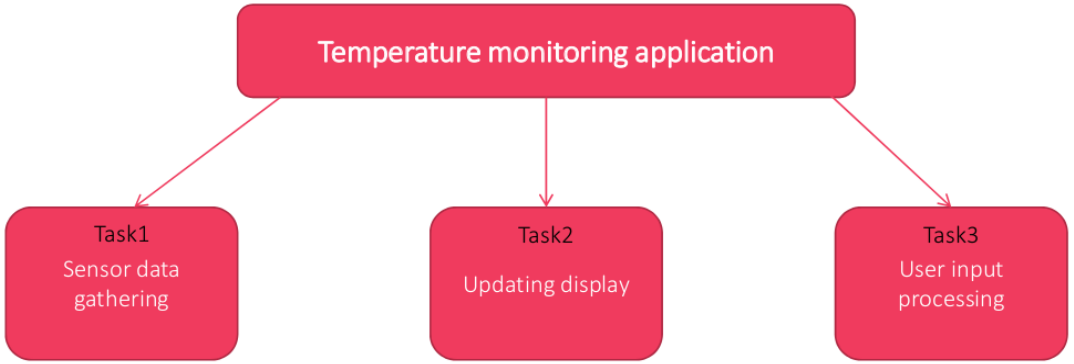
1. Priority based preemptive scheduling mechanism
2. No or very short Critical sections which disables the preemption
3. Priority inversion avoidance
4. Bounded Interrupt latency
5. Bounded Scheduling latency , etc.

RTOS

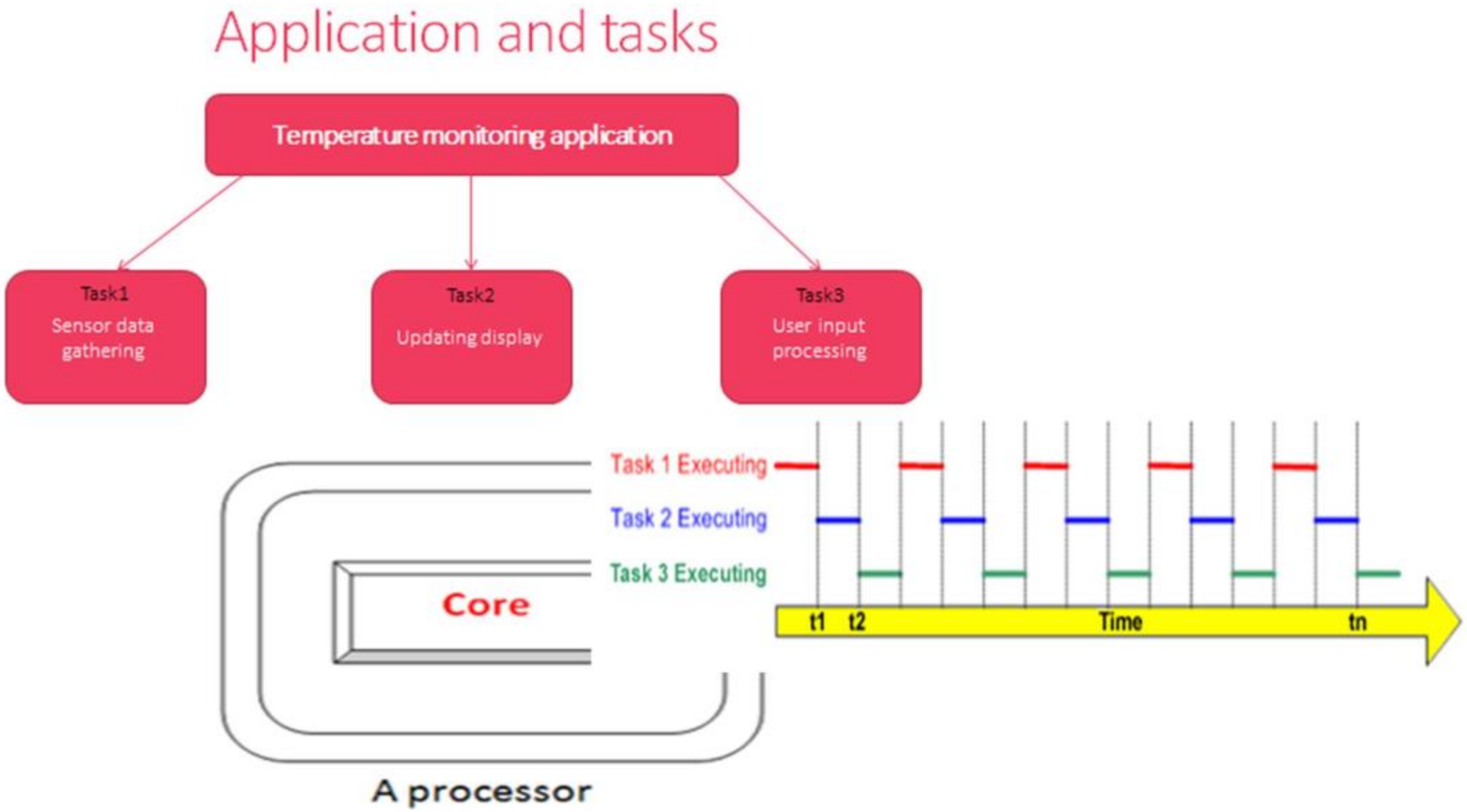
What is Multi-Tasking ?



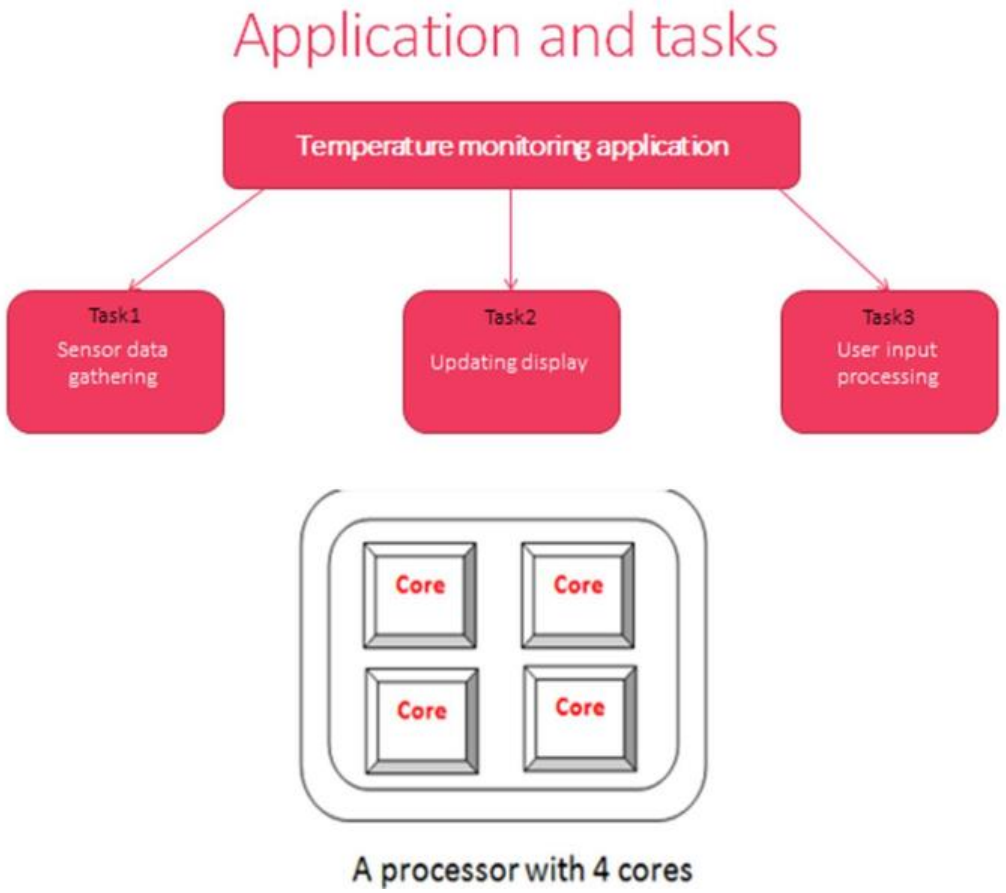
Application and tasks

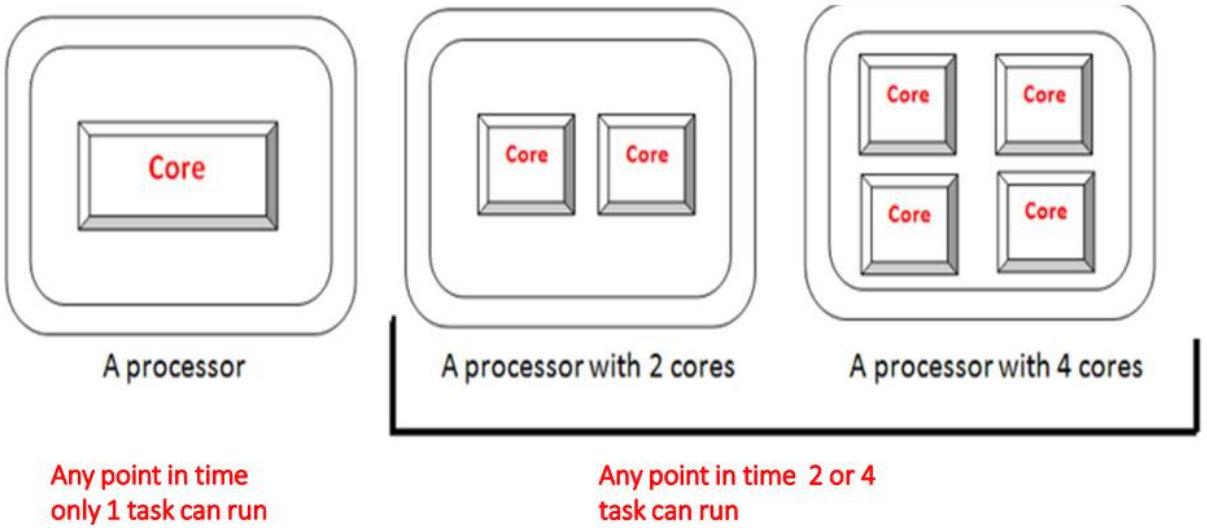


RTOS


Application and tasks
Temperature monitoring application
Task1
Sensor data gathering
Task2
Updating display
Task3
User input processing
Task 1 Executing
Task 2 Executing
Task 3 Executing
Core
t1
t2
Time
tn
A processor

RTOS


Application and tasks
Temperature monitoring application
Task1
Sensor data gathering
Task2
Updating display
Task3
User input processing
Core
Core
Core
Core
A processor with 4 cores


Core
A processor
Core
Core
A processor with 2 cores
Core
Core
Core
Core
A processor with 4 cores
Any point in time only 1 task can run
Any point in time 2 or 4 task can run

Thank You

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